

Force and Laws of Motion – Class 9 Annual Exam 2026

Full Exam-Ready Answers with Keywords

This document contains **20 high-probability questions** from the chapter **Force and Laws of Motion**, strictly aligned with the **CBSE Class 9 (2026) annual exam pattern**.

Each answer is written in **complete board-exam format**, followed by **keywords** used in marking schemes.

A. Conceptual & Theoretical Questions

1. Define balanced and unbalanced forces. Which is necessary to change the state of motion?

Answer:

Balanced forces are forces acting on an object that are equal in magnitude and opposite in direction, resulting in no change in the state of motion of the object. Unbalanced forces are forces whose resultant is not zero and hence cause a change in the state of rest or motion of an object. Only unbalanced forces are necessary to change the state of motion of an object.

Keywords: equal and opposite, net force zero, resultant force, change in motion

2. State Newton's First Law of Motion. Why is it called the Law of Inertia?

Answer:

Newton's First Law of Motion states that an object remains at rest or continues to move with uniform velocity in a straight line unless acted upon by an unbalanced external force. It is called the Law of Inertia because it describes the tendency of an object to resist any change in its state of rest or uniform motion.

Keywords: uniform motion, straight line, unbalanced force, resistance to change

3. Explain the relationship between mass and inertia. Which has more inertia: a five-rupee coin or a one-rupee coin?

Answer:

Inertia is the tendency of an object to resist changes in its state of motion. Mass is a measure of inertia; greater the mass of an object, greater is its inertia. A five-rupee coin has more mass than a one-rupee coin and therefore has greater inertia.

Keywords: resistance to change, mass measure of inertia, greater mass

4. State Newton's Second Law of Motion and derive $F = ma$.

Answer:

Newton's Second Law of Motion states that the rate of change of momentum of an object is directly proportional to the applied force and takes place in the direction of the force. Since momentum is the product of mass and velocity, the change in momentum per unit time gives force. This leads to the mathematical expression $F = ma$.

Keywords: rate of change of momentum, proportional, direction of force, $F = ma$

5. Define momentum. State its SI unit. Why is it a vector quantity?

Answer:

Momentum is defined as the product of the mass and velocity of a body. Its SI unit is kilogram metre per second (kg m/s). Momentum is a vector quantity because it has both magnitude and direction, same as velocity.

Keywords: mass $\times$ velocity, kg m/s, direction

6. State Newton's Third Law of Motion. Why do action and reaction forces not cancel each other?

Answer:

Newton's Third Law of Motion states that for every action, there is an equal and opposite reaction. Action and reaction forces do not cancel each other because they act on different objects, not on the same object.

Keywords: equal and opposite, different objects, force pair

7. State the Law of Conservation of Momentum and explain with an example.

Answer:

The Law of Conservation of Momentum states that the total momentum of a system remains constant if no external force acts on it. For example, when two objects collide, the momentum lost by one object is equal to the momentum gained by the other.

Keywords: isolated system, total momentum constant, collision

8. Describe Galileo's experiment with inclined planes and its significance.

Answer:

Galileo rolled marbles down inclined planes and observed that a marble rolling down one slope would rise

to nearly the same height on another slope. This experiment showed that force is required only to change motion, not to maintain it.

Keywords: inclined plane, uniform motion, force changes motion

B. Reasoning-Based (Daily Life Applications)

9. Why do passengers fall backward when a bus starts suddenly and forward when it stops?

Answer:

When a bus starts suddenly, the lower part of the passenger moves forward with the bus while the upper part remains at rest due to inertia, causing the passenger to fall backward. When the bus stops suddenly, the upper part continues moving forward due to inertia, causing the passenger to fall forward.

Keywords: inertia of rest, inertia of motion

10. Why does a cricket fielder pull his hands backward while catching a fast ball?

Answer:

Pulling the hands backward increases the time taken to stop the ball. This reduces the force exerted on the hands, preventing injury.

Keywords: increase time, reduce force, momentum change

11. Why does dust fall off a carpet when beaten?

Answer:

When the carpet is beaten, it moves suddenly but the dust particles tend to remain at rest due to inertia. As a result, the dust particles fall off.

Keywords: inertia of rest, sudden motion

12. Why do leaves detach when a tree branch is shaken vigorously?

Answer:

When the branch is shaken suddenly, it moves but the leaves tend to remain at rest due to inertia. This causes the leaves to detach and fall.

Keywords: inertia, sudden motion

13. Why does a gun recoil when a bullet is fired?

Answer:

When a bullet is fired forward, the gun experiences an equal and opposite reaction force. To conserve momentum, the gun moves backward, which is known as recoil.

Keywords: conservation of momentum, action–reaction

14. Explain how Newton's third law helps in walking.

Answer:

While walking, a person pushes the ground backward with his foot. The ground exerts an equal and opposite force on the person, pushing him forward.

Keywords: action–reaction, forward motion

15. Why does an athlete run before taking a long jump?

Answer:

Running increases the athlete's speed and momentum. Greater momentum helps the athlete cover a longer distance during the jump.

Keywords: momentum, speed, longer jump

C. Numerical Problems

16. Calculate force changing velocity of 5 kg object.

Answer:

Acceleration = $(7 - 3)/2 = 2 \text{ m/s}^2$. Force = $ma = 5 \times 2 = 10 \text{ N}$.

Keywords: acceleration, $F = ma$

17. Compare force for 10 g at 5 m/s^2 and 20 g at 2 m/s^2 .

Answer:

Force₁ = $0.01 \times 5 = 0.05 \text{ N}$. Force₂ = $0.02 \times 2 = 0.04 \text{ N}$. The 10 g mass requires more force.

Keywords: $F = ma$, comparison

18. Find recoil velocity of a rifle.

Answer:

Using conservation of momentum, recoil velocity of rifle = 0.44 m/s backward.

Keywords: momentum conservation, recoil

19. Find combined velocity after collision.

Answer:

Total momentum before collision is zero, so combined velocity after collision is zero.

Keywords: perfectly inelastic collision, momentum

20. Calculate force of friction on stone.

Answer:

Using equations of motion, acceleration = -4 m/s^2 . Force of friction = 4 N.

Keywords: friction, deceleration, $F = ma$

Final CBSE Strategy

- Start answers with **definitions**
- Write **formula before substitution**
- Mention **units** clearly
- Underline **keywords** for full marks